



Process Visualization in Biology: A Programming Framework for Systematic Analysis of Complex Systems

The Genome Logic Modeling Project: A Call for Experimental Validation

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Abstract

We present the Genome Logic Modeling Project (GLMP), a comprehensive computational framework that transforms biological systems analysis through standardized visual programming. Building on the evolution of Mermaid Markdown format from software documentation to scientific visualization, GLMP demonstrates that 507 biological processes across 7 kingdoms of life can be systematically represented as computational flowcharts with universal logic patterns. This work establishes a new paradigm for biological systems analysis, revealing that complex biological behaviors emerge from well-defined algorithmic processes that can be visualized, analyzed, and engineered using computational principles. We propose four testable hypotheses to validate the Programming Framework through experimental synthesis of biological circuits, supported by concrete experimental designs and validation protocols. The framework reveals universal computational patterns across viral, bacterial, and eukaryotic systems, suggesting fundamental principles of biological computation that can be validated through synthetic biology experiments.

Keywords: Computational biology, Systems biology, Synthetic biology, Biological visualization, Programming framework, Mermaid Markdown, Experimental validation

Introduction

The challenge of representing complex biological systems has long been a fundamental limitation in computational biology. Traditional approaches rely on static diagrams, mathematical models, or text-based descriptions that fail to capture the dynamic, computational nature of biological processes. The emergence of Mermaid Markdown format in 2014, originally designed for software documentation, has unexpectedly provided the foundation for a revolutionary